Mobile Robotics - Navigation of the RosBot in Webots - Group R

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Abstract

This report details the systematic development and practical implementation of a navigation system for the RosBot, operating within the Webots simulation environment. The core objective is to enable the robot to autonomously traverse its environment, proceeding from an arbitrary starting location to a designated blue pillar, and subsequently to a yellow pillar. The paramount goal throughout this process is to minimize the total simulation time required for completion. The simulated environment poses several navigational challenges, including constricted passages, areas explicitly designated as "too narrow" for traversal, and sections marked by red walls, which signify impassable zones or dead ends. Our approach integrates an occupancy grid for comprehensive environmental mapping, a robust A* pathfinding algorithm for efficient global navigation, and an edge gapping - frontier exploration strategy to discover unknown areas. The robot's precise global pose is readily available through the Webots supervisor. The system's performance is critically evaluated based on key metrics such as elapsed simulation time and the occurrence of collisions during operation.

1. Introduction

The domain of autonomous robotics continues to experience significant advancements, with navigation remaining a fundamental and complex challenge. This project focuses specifically on creating a dependable navigation solution for a mobile robot, the RosBot, within a simulated setting provided by Webots. The specific task mandates that the robot navigate from any initial position to a blue pillar, followed by navigation to a yellow pillar, with the explicit aim of achieving the shortest possible overall simulation time.

The environment itself is designed with dynamic elements and various obstacles. These include passages of varying widths, some deemed impassable ("too narrow" passages), and segments delineated by red walls, which serve as explicit indicators of dead ends or non-traversable areas. Navigating such environments demands an integration of real-time perception, environment mapping, and intelligent decision-making.

To address these challenges, we leverage a probabilistic occupancy grid approach [1], which allows the robot to build and refine a map of the environment using noisy sensor data. Frontier-based exploration strategies are employed to drive the robot toward unknown areas efficiently [3], while adaptive cost-based planning mechanisms are used to ensure safe and optimal path selection [5]. Our system also takes inspiration from efficient, scalable mapping techniques for high-resolution environments [4], and integrates goal-driven planning strategies guided by Gaussian-process-based map optimization principles [2].

A strict requirement of this project is that all solutions must be self-implemented, adhering rigorously to academic integrity standards, including appropriate citation of external references. The ultimate success of the project is evaluated based on the practical performance of the robot, including the total elapsed time, the number of collisions, and the overall effectiveness of task completion, along with a critical evaluation of the underlying algorithmic choices.

2. Literature Review

The development of autonomous mobile robots capable of exploring unknown environments, avoiding obstacles, and identifying objects of interest has been a long-standing area of research in robotics. Central to this challenge is the robot's ability to build reliable maps, reason about uncertainty, and make decisions that balance exploration with task goals. Our system draws upon several foundational and recent advances in this domain, particularly in occupancy grid mapping, frontier-based exploration, and uncertainty-aware planning.

2.1. Foundations of Occupancy Mapping

A significant milestone in robotic perception was the introduction of the probabilistic occupancy grid by Elfes [1]. This representation models the world as a grid of cells, each associated with a probability indicating the likelihood of being occupied by an obstacle. Unlike deterministic models, the occupancy grid embraces sensor uncertainty, allowing for continuous updates as new data is acquired. This framework remains the cornerstone of modern robotic mapping, and our implementation directly builds upon this concept by maintaining a confidence value for each grid cell based on LiDAR observations. We incrementally update this grid using Bayesian-style updates, marking cells as free or occupied based on beam endpoints and their traversal paths, while also inflating obstacle zones to account for the physical dimensions of the robot.

2.2. Improving Spatial Representation and Efficiency

While Elfes' model introduced the probabilistic structure, further enhancements have been proposed to improve mapping accuracy and computational efficiency. Jadidi et al. [2] proposed a mapping technique based on Gaussian Processes, enabling a continuous representation of the environment. Though our approach retains a discrete occupancy grid, we incorporate the spirit of their work by emphasizing adaptive updates and maintaining spatial continuity through map cleaning and frontier identification. Moreover, we borrow the principle of computational optimization from Ren et al. [4], who introduced ROG Map, an efficient, robocentric occupancy mapping system tailored for high-resolution, large-scale environments. Their approach highlighted the importance of local map focus and memory efficiency, which we integrate by using a fixed-size map centered around the robot's working region and cleaning up isolated or redundant data to ensure the system remains responsive.

2.3. Exploration Through Frontiers

Exploration is a fundamental behavior for autonomous robots in unknown terrain, and frontier-based strategies have proven to be both intuitive and effective. The concept, as utilized by Topiwala et al. [3], relies on identifying the boundaries between known and unknown regions of the environment and directing the robot to these "frontiers" to expand its knowledge. This approach aligns well with our system's objective of environment discovery and color-target search. In our implementation, we detect and cluster frontier cells, prune isolated candidates, and compute costmaps that guide the robot towards the most promising regions. By clustering frontiers and prioritizing them based on a combination of spatial cost and proximity to known targets (e.g., blue and yellow pillars), we ensure that exploration is not only

broad but also goal-directed.

2.4. Uncertainty and Decision-Making in Exploration

Beyond mere mapping and frontier targeting, a key consideration in modern exploration systems is the intelligent handling of uncertainty. Kim et al. [5] addressed this by introducing an uncertainty-aware exploration strategy that enhances planning efficiency by favoring actions that reduce environmental ambiguity. Inspired by their approach, our robot incorporates uncertainty into its decision-making pipeline by penalizing paths that traverse unknown regions and by computing additional clearance-based costs near obstacles. These measures result in safer, more informative trajectories and allow the robot to actively balance the trade-off between risk and reward during navigation.

2.5. Integrating Theory and Practice

Taken together, these studies inform and shape the theoretical foundation of our robot's design. From Elfes' early work on probabilistic spatial modeling to more recent advances in information-theoretic exploration and map optimization, the progression of the field is clearly reflected in our system architecture. We combine these insights to build a robust, adaptable robot capable of autonomously navigating complex spaces, identifying mission-critical objects, and continuously updating its understanding of the world in real-time. This integration of well-established techniques with modern refinements results in a system that is not only functional but also grounded in current research best practices.

3. Algorithm and Methodology

Our autonomous robot navigation system is designed to operate in unknown environments by incrementally building a map, intelligently selecting exploration goals, and detecting specific colored landmarks using onboard LiDAR and camera sensors. The system integrates real-time sensor fusion, probabilistic reasoning, adaptive motion planning, and a finite state machine to guide mission execution. This section provides an in-depth explanation of the methods and algorithms implemented, along with the motivations behind each component.

3.1. Environmental Perception and Mapping

3.1.1. Probabilistic Occupancy Grid Mapping

At the core of our robot's spatial understanding is a probabilistic occupancy grid, a concept pioneered by Elfes [1]. This approach models the environment as a two-dimensional grid, where each cell holds a probability indicating how likely it is to be occupied by an obstacle. This probabilistic model is particularly well-suited for dealing with sensor uncertainty, making it ideal for LiDAR-based navigation in unknown environments.

Initially, all cells in the grid are assigned a prior probability value of $P_{\text{UNKNOWN}} = 0.5$, representing complete uncertainty. As the robot gathers sensor data, these values are updated to reflect increasing confidence about whether a space is free or occupied.

3.1.2. Initialization and Fixed Free Space

Upon startup, the robot marks its current position and a surrounding circular area as permanently free space. This ensures that future sensor noise or mapping updates cannot falsely label the robot's starting position as an obstacle, which could hinder navigation. This "fixed" region serves as a stable foundation for building the map.

3.1.3. LiDAR Data Integration

LiDAR is the robot's primary sensor for detecting the structure of the environment. Each LiDAR beam is analyzed and processed using Bresenham's line algorithm to determine which cells it intersects. The algorithm helps update the map in two key ways:

1. **Free space update:** All cells traversed by a beam (except the endpoint) are assumed to be unoccupied and their occupancy confidence is decreased by CONFIDENCE_DECREMENT_FREE = 0.05.
2. **Obstacle detection:** The final cell hit by the beam is considered a potential obstacle and its occupancy confidence is increased by CONFIDENCE_INCREMENT_HIT = 0.08. If this confidence exceeds the threshold OCCUPIED_THRESHOLD = 0.8, it is marked as fully occupied.
3. **Neighborhood reinforcement:** Adjacent cells near confirmed obstacles receive minor confidence boosts to account for their possible spatial extent.

This method provides a probabilistic and continually refined representation of the environment based on real-time sensor data.

3.1.4. Robot Inflation for Safety:

To ensure safe path planning, the robot does not simply avoid directly occupied cells; it also maintains a safety buffer around them. A circular region with radius $R = 0.149\,\text{m}$ (equal to the robot's radius) is inflated around each confirmed obstacle, and the inflated cells are assigned a confidence value of 0.7. These are treated as non-traversable by the planner, effectively enlarging the perceived size of obstacles to reduce the risk of collision.

3.1.5. Map Cleaning and Maintenance:

As with any real-world system, sensor readings can introduce noise, leading to false positives or isolated obstacle detections. To address this, we implement two cleaning strategies:

1. **Isolated pixel cleaning:** Any obstacle cell that has fewer than five neighboring obstacles is likely a sensor artifact and is reverted to the unknown state.
2. **Frontier pruning:** Frontier cells, those located at the boundary of known and unknown regions, are essential for exploration. However, isolated frontier cells that do not contribute to meaningful exploration are removed to keep the map focused and efficient [3].

3.1.6. Camera-Based Color Detection and Localization

While LiDAR handles geometric mapping, the robot's RGB and depth cameras are used to detect and locate colored pillars—specifically red, blue, and yellow which serve as navigation targets. This capability supports mission objectives such as landmark localization or task completion.

3.1.7. Color Processing Pipeline

The color detection pipeline consists of:

1. **HSV Conversion:** RGB images are converted to HSV color space, which is more robust to lighting variations and ideal for segmenting specific colors.
2. **Masking and Contour Extraction:** Binary masks are generated for each target color using predefined HSV ranges. OpenCV's contour detection algorithm identifies potential color blobs, filtering out noise using a minimum area threshold of 20 pixels.
3. **3D Localization:** Each valid contour's center is projected into world coordinates using the depth image and camera intrinsics. This enables the robot to reason about where the object is in the global frame, not just in its field of view.

3.1.8. Color Point Cleaning and Obstacle Alignment:

To maintain reliability over time, we perform:

1. **Isolated point cleaning:** Color detections without nearby neighbors are removed to prevent chasing false positives.
2. **Alignment to obstacles:** Color points that do not coincide with nearby map obstacles are realigned to the closest occupied cells, ensuring physical consistency between vision and spatial mapping.

3.1.9. Dynamic Red Wall Extension

A unique mechanism is applied for red-colored points: once detected, a virtual "wall" is extended from the red point towards the robot's position using Bresenham's algorithm. The line—and a small circular zone at its endpoint—are marked with inflated obstacle values. This discourages the robot from entering potentially dangerous or restricted areas indicated by red markings.

3.2. Frontier-Based Exploration and Goal Selection

Exploration in unknown terrain is guided by the concept of frontier-based navigation, where the robot targets the boundaries between known free space and unexplored regions [3].

3.2.1. Frontier Detection and Clustering

Frontier cells are detected by examining map gradients, and those with adjacent unknown neighbors are marked as exploration candidates. To improve efficiency, we cluster contiguous frontier cells using a flood-fill algorithm. This allows the robot to reason about larger regions instead of individual cells.

3.2.2. Costmap Generation

Inspired by the work of Jadidi et al. [2] and Ren et al. [4], we generate a costmap using a Dijkstra-based algorithm. Costs are propagated from frontier cells across the map, assigning lower values to more accessible regions and higher costs to areas farther from the frontiers or close to obstacles.

3.2.3. Goal Prioritization

For each frontier cluster, a composite cost is computed:

$$\text{Total Cost} = \text{Dijkstra Cost} + w_d \cdot \text{Distance to Pillars} + w_u \cdot \text{Unknown Penalty} \quad (1)$$

where w_d and w_u are weights tuned to encourage goal selection near known color pillars (if detected), and to prefer frontiers adjacent to unknown space, respectively.

3.3. Adaptive and Safe Path Planning

Navigation from the current position to a selected goal is handled using a modified A* path planning algorithm. While the classic A* focuses purely on distance, we enhance it to consider uncertainty and proximity to obstacles, in line with the uncertainty-aware exploration strategies proposed by Kim et al. [5].

1. **Unknown penalty:** Paths through unknown cells incur additional cost, reflecting the risk of navigating into unobserved terrain.
2. **Clearance cost:** Cells close to obstacles are penalized based on their distance, pushing the robot to select safer routes with greater clearance.
3. **Emergency clearance:** If no valid path is found (e.g., the robot is trapped), inflated or unknown cells near the robot are temporarily treated as free to allow escape.

This planning strategy allows the robot to make context-sensitive decisions that prioritize both safety and exploration efficiency.

3.4. Safe Target Approach and Mission Completion

Upon detecting a blue or yellow pillar, the robot computes a waypoint that lies at a safe standoff distance of $R + 0.1$ m from the pillar. If this position is not navigable, the planner searches nearby for the nearest valid alternative.

The mission is orchestrated through a finite state machine with five states: `INITIAL_SCAN`, `EXPLORE`, `GO_TO_BLUE`, `GO_TO_YELLOW`, and `MISSION_COMPLETE`. Initially, the robot performs a 360-degree scan to detect any visible pillars. It then begins exploring the map, navigating toward frontiers and prioritizing movement toward detected pillars. Once both blue and yellow targets are reached within 0.35 m, the robot concludes its mission and halts.

4. Results and Discussion

To evaluate the effectiveness of our autonomous robot navigation system, we conducted a series of experiments across five distinct maze environments, labeled Maze 1 through Maze 5. These mazes vary in terms of complexity, layout density, obstacle distribution, and visual clutter, thereby offering a comprehensive test of our system's robustness, adaptability, and perception accuracy.

4.1. Performance in Maze Environments

4.1.1. Mazes 3, and 5 — Successful Operation , Maze 1 is partially done

In Maze 3, and Maze 5, the robot consistently demonstrated reliable and stable autonomous behavior. Maze 1 was only half, because it couldnt find path to a valid destination within a given radius But by slightly increasing the search radius the rosbot wouldve went near yellow pillar, Across these environments, the following outcomes were observed:

1. **Accurate Mapping:** The occupancy grid was progressively and accurately constructed using LiDAR data, with minimal artifacts. Free and occupied zones were well differentiated.
2. **Effective Exploration:** Frontier-based exploration enabled the robot to reach full coverage of the map, even in the presence of non-linear walls and dead-ends. The Dijkstra-based costmap and A* planner efficiently guided the robot around obstacles with adequate clearance.
3. **Robust Pillar Detection:** Blue and yellow pillars were successfully detected and localized using the RGB-D camera pipeline. The HSV-based color segmentation and 3D coordinate projection performed reliably, even under variable lighting.
4. **Safe Navigation and Collision Avoidance:** The robot maintained safe paths due to the effective inflation of obstacles. In all successful mazes, the robot avoided collisions with both static and dynamically interpreted red zones.
5. **Mission Completion:** The robot transitioned through all mission states from initial scanning to successful interaction with both colored pillars, culminating in correct and timely mission termination.

These results validate the system's core functionalities and confirm that the integration of probabilistic mapping, adaptive path planning, and color-based object detection is effective in structured, moderately complex environments.

4.1.2. Mazes 2 and 4 — Partial Success and Challenges

In Maze 2 and Maze 4, the robot exhibited some operational difficulties that occasionally hindered full mission completion:

1. **Mapping Disruptions:** In these mazes, which featured narrower passages and irregular obstacle placements, the occupancy grid sometimes developed inconsistent frontier regions or false obstacles. This was primarily due to increased LiDAR occlusion and sensor noise near tight corners.
2. **Exploration Inefficiency:** The frontier clustering and selection algorithm, while generally robust, occasionally favored isolated or less-informative regions, leading to suboptimal exploration paths and increased traversal time.
3. **Unstable Color Detection:** In Maze 4, excessive lighting and environmental reflections affected the accuracy of the HSV-based color masking. This caused transient or missed detections of colored pillars, which impeded transitions into the goal-oriented states.
4. **Path Planning Failures:** In some cases, inflated obstacle regions expanded into narrow corridors, making valid pathfinding impossible. Although the emergency clearance mechanism was triggered, it was not always sufficient to recover navigation.

Despite these issues, the robot was still able to partially explore the environment and, in some runs, reach at least one of the two pillars.

These challenges highlight potential areas for further enhancement, particularly in dynamic inflation tuning, more adaptive frontier evaluation, and robustifying color detection against environmental noise.

4.2. System Strengths and Limitations

The experimental results illustrate several key strengths of our approach:

1. **Generalization Across Mazes:** The robot performed well in structurally diverse environments, indicating good generalizability of the exploration and mapping framework.
2. **Integrated Perception and Planning:** The coupling of vision-based color detection with probabilistic mapping allowed high-level reasoning about exploration goals.
3. **Safety-Aware Navigation:** The robot consistently avoided collisions, demonstrating the effectiveness of clearance-based path planning.

However, the experiments also reveal limitations:

1. **Sensitivity to Sensor Noise:** In complex geometries, noise from LiDAR or lighting variations in the camera affected mapping and detection reliability.
2. **Over-conservative Obstacle Inflation:** In narrow environments, the inflated zones sometimes restricted pathfinding more than necessary.
3. **Static Color Thresholds:** Fixed HSV ranges limited robustness to illumination changes. Adaptive color calibration could improve performance in real-world deployments.

4.3. Discussion

Overall, the system demonstrates strong potential for autonomous indoor navigation and target detection in controlled environments. The results affirm the utility of integrating frontier-based exploration with uncertainty-aware planning and multimodal sensing. Moving forward, improvements such as real-time adaptive thresholding, machine learning-based color classification, or dynamic map tuning could address the observed limitations and expand the system's applicability to even more challenging scenarios.

The developed autonomous robot system successfully demonstrates the integration of real-time perception, robust mapping, intelligent path planning, and a state-driven mission control. It effectively explores unknown environments, accurately identifies and approaches specific colored objectives, and adapts to environmental changes. While current limitations provide clear directions for future enhancements, the system establishes a strong foundation for complex autonomous navigation tasks.

5. Limitations and Challenges

While the developed autonomous navigation system demonstrated strong performance in several maze environments, the project also revealed a number of limitations and operational challenges that constrain its current capabilities and highlight opportunities for future improvement.

5.1. 1. Sensor Noise and Perception Reliability

One of the primary limitations observed was the system's sensitivity to sensor noise, particularly in cluttered or narrow environments. In Maze 2 and Maze 4, false obstacle detections occasionally resulted from LiDAR reflections off nearby walls or from edge cases in the depth camera data. This noise occasionally led to incomplete or distorted occupancy maps, which impaired navigation.

Similarly, the RGB camera's color detection pipeline, while effective under controlled lighting, struggled under harsh or inconsistent illumination. Shadows, reflections, and color bleeding reduced the reliability of red, blue, and yellow pillar detection. The use of fixed HSV thresholds made the system susceptible to these environmental changes.

5.2. 2. Over-Conservative Obstacle Inflation

To maintain safe navigation, the robot applies a fixed inflation radius around detected obstacles. However, in tighter environments such as Maze 4, this safety margin sometimes caused navigable paths to appear blocked, resulting in pathfinding failures or unnecessary detours. The inability to dynamically adjust inflation based on the context or environment layout reduced efficiency and occasionally prevented goal completion.

5.3. 3. Exploration Inefficiencies

The frontier-based exploration strategy generally guided the robot effectively. However, in some cases, the robot selected small or isolated frontiers that contributed little to overall map coverage. This led to inefficient paths or exploration dead-ends. Moreover, frontier clustering and ranking did not always prioritize the most informative or goal-relevant regions, especially when color pillars were detected in distant or partially mapped zones.

5.4. 4. Recovery from Deadlocks and Traps

While the system includes an emergency clearance mechanism to escape local traps, it remains relatively simplistic. In situations where the robot became surrounded by inflated or uncertain cells, the recovery strategy did not always find a viable path forward, occasionally resulting in the robot stalling or oscillating without progress. This was more evident in mazes with narrow passages and minimal clearance.

5.5. 5. Static Color Thresholding

The use of static HSV thresholds for color detection poses a fundamental limitation in dynamic environments. Variations in lighting, angle, and camera exposure significantly affected detection accuracy. The system lacks adaptive calibration or learning-based classification, which limits its applicability in uncontrolled or variable lighting conditions.

5.6. 6. Limited Generalization to Real-World Environments

While the system performed well in simulation and structured maze setups, it has not yet been validated in real-world settings with irregular floor textures, dynamic obstacles, and non-ideal sensor conditions. Bridging this sim-to-real gap would require significant enhancements in robustness, error handling, and system calibration.

These challenges underscore the complexity of autonomous navigation in unknown environments. They highlight the importance of adaptive perception, contextual decision-making, and robust error recovery mechanisms. Addressing these limitations in future work will significantly enhance the system's autonomy, reliability, and applicability in real-world scenarios.

6. Conclusion

In this project, we developed and implemented an autonomous robotic navigation system capable of exploring unknown environments, constructing accurate occupancy maps, and detecting colored target pillars using LiDAR and RGB-D camera inputs. By integrating probabilistic occupancy grid mapping, frontier-based exploration, adaptive path planning, and robust color-based object detection, the robot was able to navigate complex indoor mazes with minimal human intervention.

The system demonstrated reliable and consistent performance in multiple test environments, notably completing missions successfully in Maze 3, 5 and Maze 1 was only half, because it couldnt find path to a valid destination within a given radius But by slightly increasing the search radius the rosbot wouldve went near yellow pillar. These results validate the effectiveness of our mapping and navigation strategy, particularly the fusion of LiDAR-based occupancy mapping with real-time decision-making driven by frontier analysis and color-based goal prioritization. The robot maintained safe

navigation paths through clearance-aware planning, and its mission control architecture allowed smooth transitions between exploration and goal-seeking behaviors.

However, challenges encountered in Maze 2 and Maze 4 highlighted key areas for improvement, including sensitivity to sensor noise in tight environments, the limitations of static color thresholds under variable lighting, and the need for more adaptive obstacle inflation strategies. These insights pave the way for future enhancements.

Looking ahead, the system could be further improved by incorporating adaptive color calibration, learning-based perception modules, and more dynamic frontier evaluation strategies. Additionally, extending the framework to operate in larger-scale or outdoor environments would be a valuable next step toward real-world deployment.

Overall, this project demonstrates the viability of a modular, sensor-fusion-driven approach to autonomous navigation and object detection in unknown environments, contributing meaningfully to the broader field of mobile robotics.

6.1. Future Work

1. Improvement in Simulation Time.
2. Integrate SLAM for better localization
3. Improve visual robustness under lighting changes
4. Share maps among multiple robots
5. Deploy on physical platforms

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